WORKED SOLUTION

STEP I 1999, Question 6

Pure • Extrema of Quadratics

STEP I 1999, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q6

The key idea

A straight line on a closed interval has its extreme values at the ends, and which end gives which depends only on the sign of the gradient. For a quadratic with $c > 0$ the graph is a $\cup$-shaped parabola: the greatest value is still at an end, but the least value is at the vertex *if the vertex lies in the interval*, and at the nearer end otherwise. So the whole question comes down to asking where the vertex $x = -b/(2c)$ sits relative to $[-10, 10]$.

Solution

Part (i)

Write $f(x) = bx + a$, so $f(10) = a + 10b$ and $f(-10) = a - 10b$.

- If $b > 0$, f is strictly increasing, so the least value is $f(-10) = a - 10b$ (at $x = -10$) and the greatest is $f(10) = a + 10b$ (at $x = 10$).
- If $b = 0$, $f(x) = a$ for every x, so the greatest and least values are both a.
- If $b < 0$, f is strictly decreasing, so the greatest value is $f(-10) = a - 10b$ (at $x = -10$) and the least is $f(10) = a + 10b$ (at $x = 10$).

All three cases are summed up by

$$\text{greatest} = a + 10|b|, \qquad \text{least} = a - 10|b|.$$

Part (ii)

Let $g(x) = cx^2 + bx + a$ with $c \geq 0$. If $c = 0$ this is part (i), so the answer is $a \pm 10|b|$ as above. From now on take $c > 0$ and complete the square:

$$g(x) = c\left(x + \frac{b}{2c}\right)^2 + a - \frac{b^2}{4c}.$$

The graph is a ∪-shaped parabola with vertex at $x_0 = -\dfrac{b}{2c}$, where g takes its smallest value over *all* real x. Also g decreases for $x < x_0$ and increases for $x > x_0$. The end values are

$$g(10) = a + 100c + 10b, \qquad g(-10) = a + 100c - 10b.$$

Greatest value.

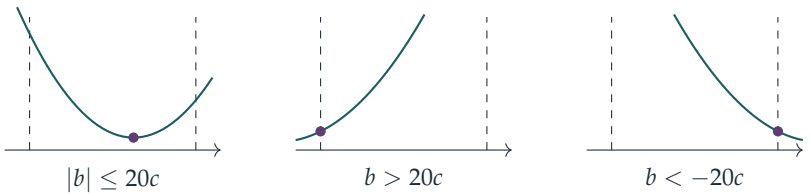
Ask yourself Why must the greatest value of g on the interval be at one of the ends, wherever the vertex is?

Because $c > 0$, $g(x)$ depends on x only through the distance $|x - x_0|$, and it increases as that distance increases. On $[-10, 10]$ the distance $|x - x_0|$ is largest at one of the ends (whichever is further from x_0). So the greatest value is the larger of $g(10)$ and $g(-10)$, and these differ by $g(10) - g(-10) = 20b$. Hence:

- if $b > 0$, the greatest value is $a + 100c + 10b$, at $x = 10$;
- if $b < 0$, the greatest value is $a + 100c - 10b$, at $x = -10$;
- if $b = 0$, the greatest value is $a + 100c$, at both ends.

In every case, greatest $= a + 100c + 10|b|$. This also holds when $c = 0$.

Least value. Now it matters where x_0 is. The vertex lies in the interval exactly when $-10 \le -\frac{b}{2c} \le 10$, that is (multiplying by $2c > 0$) when $-20c \le b \le 20c$, or $|b| \le 20c$.



The dashed lines are $x = \pm 10$, and the dot marks the least value on the interval.

- $|b| \le 20c$ **(vertex in the interval).** The least value is the global minimum,

$$a - \frac{b^2}{4c}, \quad \text{at } x = -\frac{b}{2c}.$$

- $b > 20c$ **(vertex to the left, $x_0 < -10$).** On $[-10, 10]$ we have $x > x_0$, so g is increasing there and the least value is $g(-10) = a + 100c - 10b$, at $x = -10$.
- $b < -20c$ **(vertex to the right, $x_0 > 10$).** Now g is decreasing on the whole interval, so the least value is $g(10) = a + 100c + 10b$, at $x = 10$.

The two outside cases combine to give least $= a + 100c - 10|b|$ when $|b| > 20c$.

Ask yourself What happens on the boundary $|b| = 20c$? Do the two formulas clash there?

No. If $b = 20c$ then $a - \frac{b^2}{4c} = a - 100c$ and $a + 100c - 10b = a - 100c$ as well, because the vertex is then exactly at $x = -10$. The case $b = -20c$ works the same way. So the answer changes continuously as b crosses $\pm 20c$, and it doesn't matter which case includes the boundary.

Summary. For $-10 \le x \le 10$ and $c \ge 0$:

$$\text{greatest} = a + 100c + 10|b|, \qquad \text{least} = \begin{cases} a - \dfrac{b^2}{4c}, & c > 0 \text{ and } |b| \le 20c, \\ a + 100c - 10|b|, & |b| > 20c, \\ a, & b = c = 0. \end{cases}$$

The case $c = 0$, $b \neq 0$ falls under $|b| > 20c = 0$ and gives back part (i).

Checks

- Take $g(x) = x^2 + 4x + 1$: here $|b| = 4 \le 20$, so the least value is $1 - 4 = -3$ at $x = -2$, and the greatest is $1 + 100 + 40 = 141$ at $x = 10$. Direct substitution agrees.
- Take $g(x) = x^2 + 30x$: here $b = 30 > 20$, so the vertex $x = -15$ is outside the interval and the least value is $g(-10) = 100 - 300 = -200$. Indeed $g(-10) = -200$ and g is increasing on $[-10, 10]$.
- A short computer check compared the formulas with a brute-force search over 401 points in $[-10, 10]$ (exact rational arithmetic) for 1000 random (a, b, c), including $c = 0$ and the boundary $|b| = 20c$. All cases agree.

Common slips

- Quoting $a - \frac{b^2}{4c}$ as the least value without checking that the vertex lies in $[-10, 10]$.
- Forgetting the cases $c = 0$ (no vertex, and you can't divide by c) and $b = c = 0$.
- Mixing up which end gives the least value when the vertex is outside: it's the end *nearer* the vertex.

Beyond STEP

The fact behind this question is the Extreme Value Theorem: a continuous function on a closed, bounded interval attains a greatest and a least value, and these occur either at an end or at a stationary point inside. The observation that a convex function

such as $cx^2 + bx + a$ with $c \geq 0$ attains its maximum at an end is the first step in convex optimisation, which is behind much of modern applied mathematics. Abbott's *Understanding Analysis* proves the Extreme Value Theorem carefully.

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